

NY TECH WEEK

AI / ML HACKATHON
INTUIT
NEW YORK, NEW YORK | 6.5 - 6.6

51 ASTOR PLACE

#NYTECHWEEK

Welcome

- AI Data Scientists
- Hackers
- Innovators

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INTUIT

NYC

AI Hub

Fastest growth site

Top talent

Awesome perks

Prime real estate

Great culture

Agenda

Introduction

Schedule

The challenge

Prizes

Logistics

Q & A

Schedule

FRIDAY (Day 1)

Time	Activity
16:00-17:00	Doors Open + Check-in
17:00-17:30	Opening Session
17:30-18:00	Hacking
18:00	Dinner 🍕
18:00-20:00	Hack On!
20:00	Lights Out for Day 1

SATURDAY (Day 2)

Time	Activity
8:30	Doors Open + Breakfast 🍕
9:00-12:00	Hacking
12:00	Lunch 🍕
12:00-14:00	Hack On!
14:00	Submission Deadline 💀
14:00-16:00	Networking Session + Drinks 🍕
16:00-17:00	Award Ceremony + Closing Keynote
17:00-18:00	Photo-Op + Networking + More Drinks 🍕
18:00	That's a Wrap!

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The Challenge

You are a small-business lender.

Using a set of historical loan applications, decide whom to fund to maximize portfolio profit, and be able to defend why, including what would happen under interventions you never observed.

More precisely:

The lender chooses a policy $d : \mathcal{X} \rightarrow \{0, 1\}$ (approve = 1) to maximize realized portfolio value:

$$\max_d \mathbb{E} \left[\sum_i d(\mathbf{x}_i) \cdot \text{NPV}_i \right], \quad \text{NPV}_i = \begin{cases} F_i + R_i r \frac{T}{365} & y_i = 0 \text{ (repaid)} \\ F_i + D_i (t_i^* - 1) + \text{rec}_i - R_i & y_i = 1 \text{ (default at day } t_i^*) \end{cases}$$

with requested amount R_i , APR $r = 0.35$, term $T = 60$ days, daily draw $D_i = R_i(1 + rT/365)/T$, origination fee $F_i = 0.03R_i$.

To that end, we have the following deliverables:

Deliverable A - Given a set of loan applications, predict the *default probability*, y with the 90% PI of each, and act on it (*accept or reject*)

$$\hat{p}_i = \Pr(y_i = 1 \mid \mathbf{x}_i), \quad d_i = \mathbb{I}[\mathbb{E}[\text{NPV}_i \mid \text{approve}] > 0]$$

(A naive policy uses a flat threshold $d_i = \mathbb{I}[\hat{p}_i < \tau]$; the economically coherent one uses the NPV sign.)
Plus a 90% interval $[\ell_i, u_i]$ with $\ell_i \leq \hat{p}_i \leq u_i$.

Deliverable B - For each origination cohort, w and loan age, a weeks, the cumulative default fraction of the team's own approved set $\mathcal{A}_w$

$$CDR_{w,a} = \frac{|\{i \in \mathcal{A}_w : t_i \leq 7a\}|}{|\mathcal{A}_w|}$$

NOTE: A and B *can* share one model, but **B needs more** — e.g. $\Pr(\text{default by day } t)$ for all t . A day-5 default and a day-55 default look identical to A but completely different to B (and to NPV).

To that end, we have the following deliverables:

Deliverable C - Given applications with interventions, estimate their new counterfactual default probabilities, p_q^{cf}

$$p_q^{cf} = \Pr(y_{i_q} = 1 \mid \text{do}(f_q = v_q), \mathbf{X}_{i_q, -f_q} = \mathbf{x}_{i_q, -f_q})$$

$$\underbrace{\Pr(y \mid \text{do}(X = x))}_{\text{interventional (what we ask)}} \neq \underbrace{\Pr(y \mid X = x)}_{\text{observational (naive perturbation)}} \quad \text{whenever } X \text{ is confounded.}$$

Deliverable D - The simplest and most fun, a 4-page (max) write up on your approach and solution to the challenge. Template will be provided in the challenge Github repository.

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The Dataset

100K SMB loan applications over ~18 months of history + a 3-month test window

applicant_id	sector	requested_amount	obs_monthly_rev_3mo	credit_util	prior_decision	default_flag	days_to_default	...
e5a5dd55...	1	23,131	180,063	0.46	0 (declined)	—	—	...
cf18287f...	3	17,189	80,968	0.61	1 (approved)	0	—	...
9a9fe9df...	0	30,185	117,083	0.72	1 (approved)	0	—	...
561cc418...	0	29,293	null	0.70	0 (declined)	—	—	...
d09bfa55...	1	22,280	null	0.24	1 (approved)	1	43	...

8 feature groups:

- Business identity
- Bank-feed derived (partial coverage)
- Platform engagement
- Prior-underwriter outputs
- Self-reported (optimistically biased)
- Bureau-style credit
- Application context
- Outcomes (approved-only, nullable)

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The Deliverables

A
Application Decisions
B
Default Ratio wrt Time
C
Counterfactual Estimation
D
Your Writeup

Final Score

$$S = 0.30 S_{P\&L} + 0.25 S_{traj} + 0.20 S_{cal} + 0.10 S_C + 0.15 S_{write}, \quad S \in [0, 1]$$

$S_{P\&L}$: Realized portfolio value of the loans you chose to fund (Deliverable A decisions)

S_{traj} : Accuracy of your cohort default-trajectory forecast over time (Deliverable B)

S_{cal} : How well your 90% intervals on PD (A) and trajectory (B) contain the truth without being needlessly wide

S_C : How close your interventional PD predictions are to the true causal effects (Deliverable C)

S_{write} : Quality of your methodological defense (Deliverable D)

A winning team

Funds a genuinely profitable loan book

Accurately forecasts how that book defaults over time

Gives well-calibrated uncertainty intervals

Returns counterfactual PDs that match the true interventional effects (not naive re-prediction)

Defends every choice in a clear writeup

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Getting Started



[Dataset, Template, Instructions](#)



[Team Registration \(inc. Solo!\)](#)



[Hackathon Schedule](#)

ATTENTION

You must fill the [Team Registration](#) form ASAP in order to get your submission link. Please get it down by [8PM Friday](#).

No team changes are allowed after.

You will get your submission links on Saturday.

Before submitting, please [follow the README](#) and validate your submission with the provided script.